

What is FAIR and AI-Ready for Datasets?

Bhavesh Patel, Ph.D.
Research Professor



CALIFORNIA MEDICAL
INNOVATIONS INSTITUTE

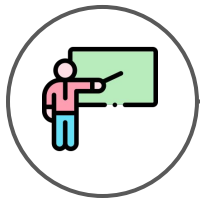


FAIR DATA
INNOVATIONS HUB

About This Presentation



20 min



How to make research
data FAIR and AI-ready?

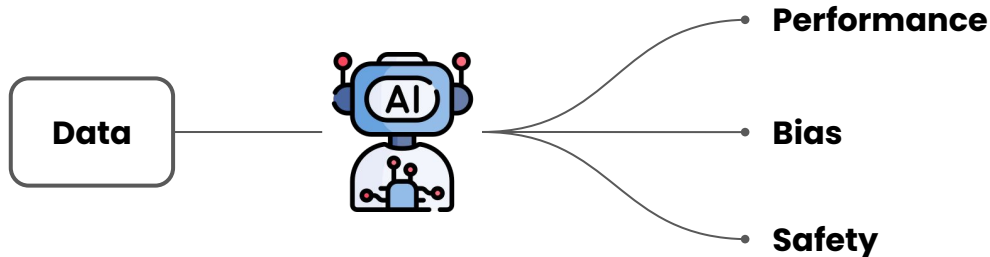


AI-Ready Data

AI-Ready Data

Why make research data ready for AI?

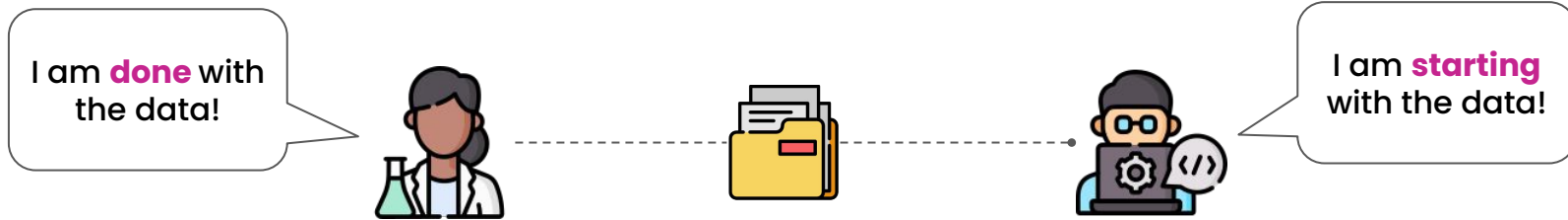
To **facilitate the development** of AI models and ensure they use data for the **right purpose** and in the **right way**



AI-Ready Data

Who is responsible for making data AI-ready?

The burden of making data AI-ready largely falls on the **data generators/creators**



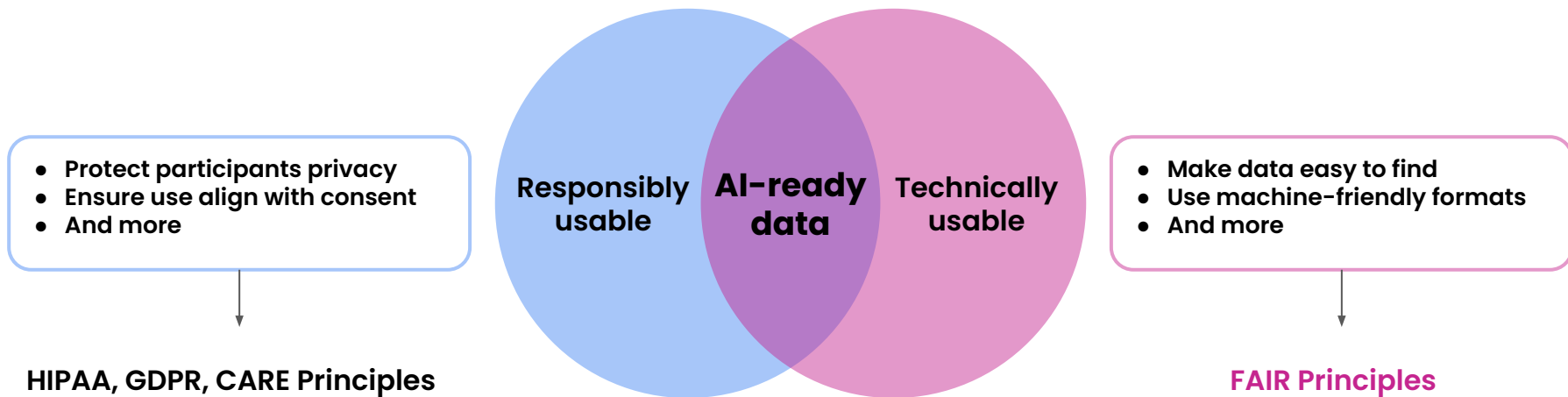
Data generators

AI developers

- Ensure your data is not misused
- Increase the impact of your data

AI-Ready Data

How to make data AI-ready?



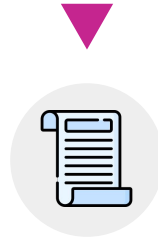


FAIR Principles

FAIR Principles

Origin

How to make all research outcomes, including data,
optimally reusable by humans and machines?



FAIR Principles (2016)

FAIR Principles

15 principles to optimize data reuse for humans and machines

To be Findable:

- F1. (meta)data are assigned a globally unique and persistent identifier
- F2. data are described with rich metadata (defined by R1 below)
- F3. metadata clearly and explicitly include the identifier of the data it describes
- F4. (meta)data are registered or indexed in a searchable resource

To be Accessible:

- A1. (meta)data are retrievable by their identifier using a standardized communications protocol
 - A1.1 the protocol is open, free, and universally implementable
 - A1.2 the protocol allows for an authentication and authorization procedure, where necessary
- A2. metadata are accessible, even when the data are no longer available

To be Interoperable:

- I1. (meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
- I2. (meta)data use vocabularies that follow FAIR principles
- I3. (meta)data include qualified references to other (meta)data

To be Reusable:

- R1. meta(data) are richly described with a plurality of accurate and relevant attributes
 - R1.1. (meta)data are released with a clear and accessible data usage license
 - R1.2. (meta)data are associated with detailed provenance
 - R1.3. (meta)data meet domain-relevant community standards

FAIR Principles

Interpretation



Findable → Data are easy to find

Data have a unique identifier (e.g. a DOI)
Metadata are provided and indexed in search engines



Accessible → Data access process is clear

Data are accessible using a standard protocol (e.g., HTTP)
Access requirements are clearly stated



Interoperable → Data work well with related data/tools

Data and metadata are in standard format
Data are connected to related resources



Reusable → Data is easily usable

Data have a clear usage license
Data are well documented

FAIR Principles

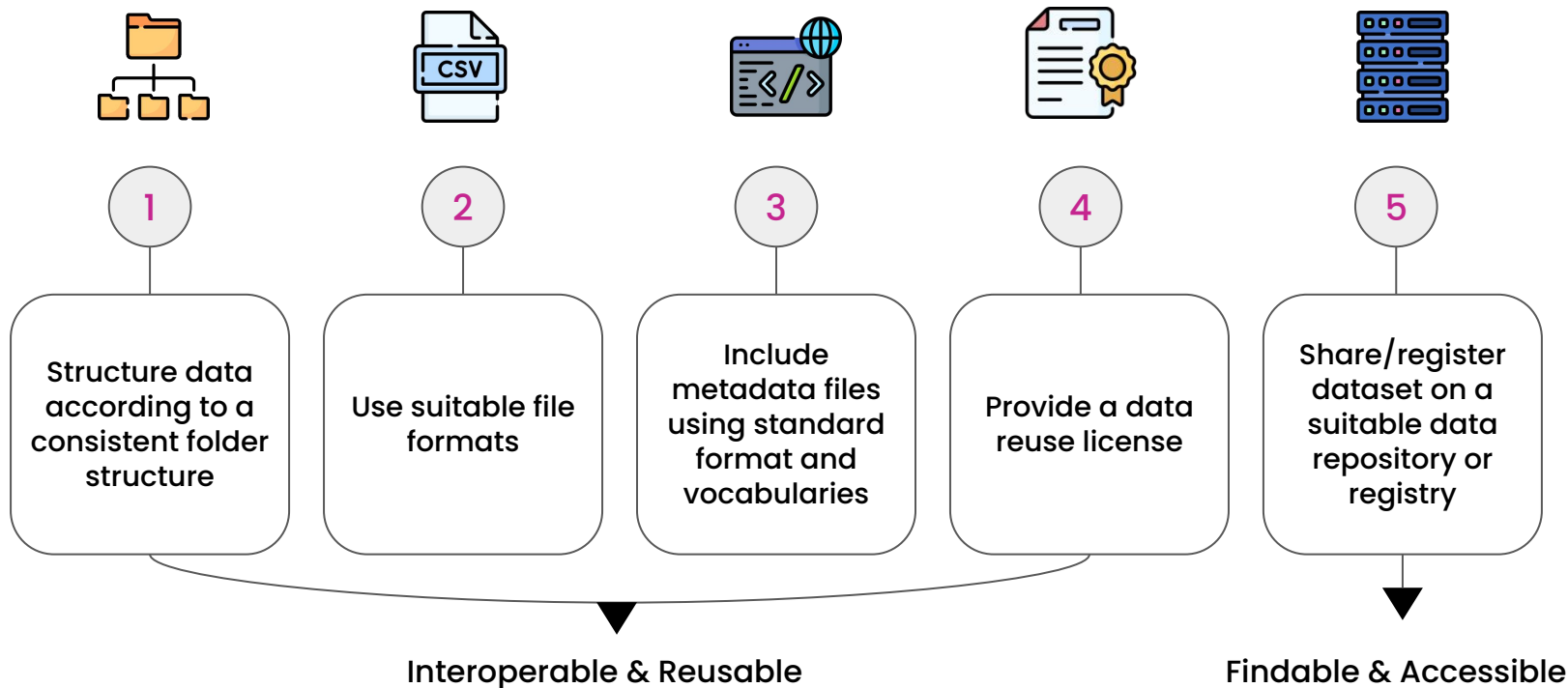
What they are not

FAIR  Open

FAIR  Ethical or fair

FAIR Principles

How to practically make data FAIR





The AI-READI Dataset

The AI-READI Dataset

About AI-READI

AI-READI is one of the **Data Generation Project (DGP)** of the
NIH Bridge2AI Common fund program

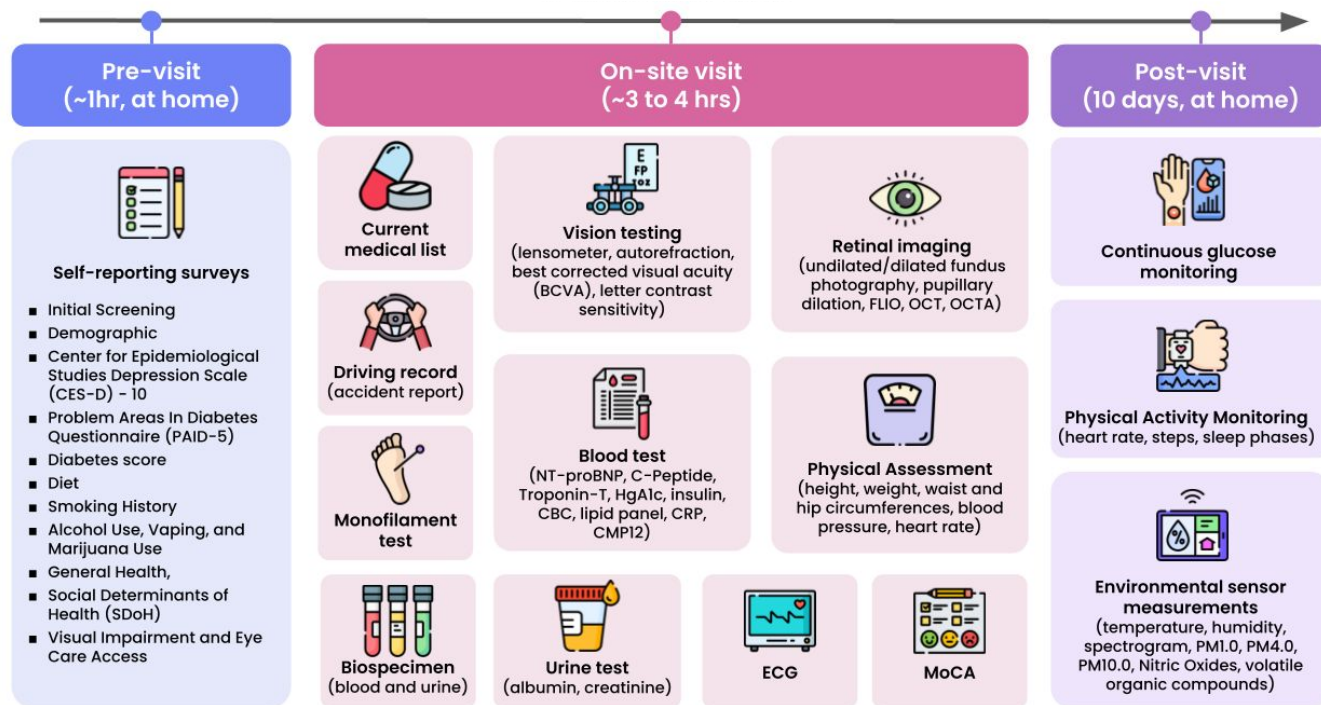


Collecting a multimodal dataset
for studying Type 2 Diabetes

<https://aireadi.org>

The AI-READI Dataset

Data collection protocol



FLO = Fluorescence Lifetime Imaging, OCT = Optical Coherence Tomography, OCTA = Optical Coherence Tomography Angiography,
ECG = Electrocardiogram, MoCA = Montreal Cognitive Assessment, PM1.0, 4.0, and 10.0 = Particulate matter less than 1, 4, and 10 microns, respectively

The AI-READI Dataset

FAIR: 1. Consistent folder structure

Clinical Dataset Structure (CDS)

cds-specification.readthedocs.io



- cardiac_ecg
- clinical_data
- environment
- retinal_flio
- retinal_oct
- retinal_octa
- retinal_photography
- wearable_activity_monitor
- wearable_blood_glucose
- CHANGELOG.md
- LICENSE.txt
- README.md
- dataset_description.json
- dataset_structure_description.json
- healthsheet.md
- participants.json
- participants.tsv
- study_description.json

One folder per
datatype

Metadata files

The AI-READI Dataset

FAIR: 2. Standard data format

Data category	Data format followed
Clinical data	Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM)
Retinal imaging data	Digital Imaging and Communications in Medicine (DICOM)
ECG data (standard 12-lead)	WaveForm DataBase (WFDB)
Wearables data (physical activity monitoring, continuous glucose monitoring data)	Open mHealth
Environmental sensor data	Earth Science Data Systems (ESDS) format

The AI-READI Dataset

FAIR: 3. Extensive metadata

retinal_photography

wearable_activity_monitor

- CHANGELOG.md
- dataset_description.json
- dataset_structure_description.json
- healthsheet.md
- LICENSE.txt
- participants.json
- participants.tsv
- README.md
- study_description.json

Broad metadata: study background, dataset provenance, dataset structure, license terms, participants information

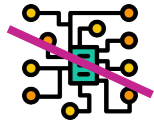
Provided in both human and machine-friendly formats

The AI-READI Dataset

FAIR: 4. Shared under a new usage license



Prohibits re-identification of study participants or harm



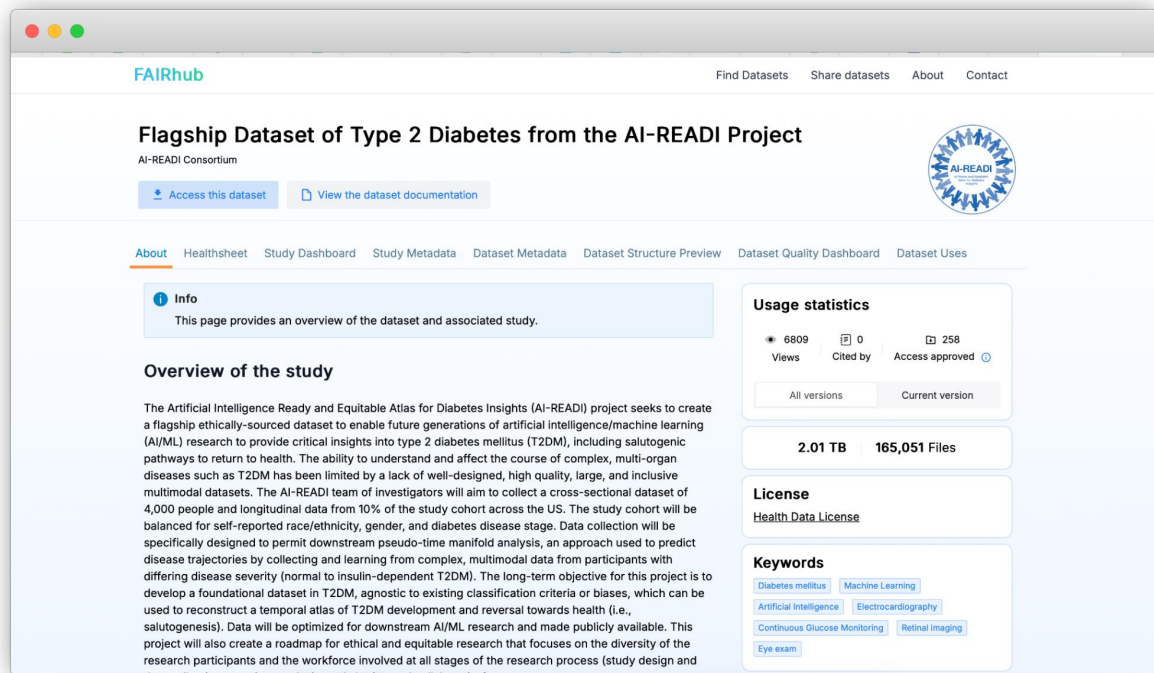
Prohibits distribution of the data and of models that may “copy” or “memorize” the data



Allows commercial use

The AI-READI Dataset

FAIR: 5. Shared on FAIRhub



The screenshot shows the FAIRhub website interface for the 'Flagship Dataset of Type 2 Diabetes from the AI-READI Project'. The page includes a navigation bar with links to 'Find Datasets', 'Share datasets', 'About', and 'Contact'. Below the title, there are buttons for 'Access this dataset' and 'View the dataset documentation'. A circular logo for the AI-READI Consortium is visible. The main content area features a tabbed interface with 'About' selected, showing an 'Info' section and an 'Overview of the study' section. The 'Overview of the study' section contains a detailed paragraph about the dataset's purpose and scope. On the right side, there are three summary boxes: 'Usage statistics' showing 6809 views, 0 citations, and 258 access approvals; a storage and file count box showing 2.01 TB and 165,051 files; and a 'License' section indicating 'Health Data License'. A 'Keywords' section at the bottom right lists terms like 'Diabetes mellitus', 'Machine Learning', 'Artificial Intelligence', 'Electrocardiography', 'Continuous Glucose Monitoring', 'Retinal imaging', and 'Eye exam'.

FAIRhub

Find Datasets Share datasets About Contact

Flagship Dataset of Type 2 Diabetes from the AI-READI Project

AI-READI Consortium

[Access this dataset](#) [View the dataset documentation](#)

[About](#) [Healthsheet](#) [Study Dashboard](#) [Study Metadata](#) [Dataset Metadata](#) [Dataset Structure Preview](#) [Dataset Quality Dashboard](#) [Dataset Uses](#)

Info
This page provides an overview of the dataset and associated study.

Overview of the study

The Artificial Intelligence Ready and Equitable Atlas for Diabetes Insights (AI-READI) project seeks to create a flagship ethically-sourced dataset to enable future generations of artificial intelligence/machine learning (AI/ML) research to provide critical insights into type 2 diabetes mellitus (T2DM), including salutogenic pathways to return to health. The ability to understand and affect the course of complex, multi-organ diseases such as T2DM has been limited by a lack of well-designed, high quality, large, and inclusive multimodal datasets. The AI-READI team of investigators will aim to collect a cross-sectional dataset of 4,000 people and longitudinal data from 10% of the study cohort across the US. The study cohort will be balanced for self-reported race/ethnicity, gender, and diabetes disease stage. Data collection will be specifically designed to permit downstream pseudo-time manifold analysis, an approach used to predict disease trajectories by collecting and learning from complex, multimodal data from participants with differing disease severity (normal to insulin-dependent T2DM). The long-term objective for this project is to develop a foundational dataset in T2DM, agnostic to existing classification criteria or biases, which can be used to reconstruct a temporal atlas of T2DM development and reversal towards health (i.e., salutogenesis). Data will be optimized for downstream AI/ML research and made publicly available. This project will also create a roadmap for ethical and equitable research that focuses on the diversity of the research participants and the workforce involved at all stages of the research process (study design and data collection, analysis, synthesis, and dissemination).

Usage statistics

6809 Views 0 Cited by 258 Access approved

All versions Current version

2.01 TB 165,051 Files

License

Health Data License

Keywords

Diabetes mellitus Machine Learning Artificial Intelligence Electrocardiography Continuous Glucose Monitoring Retinal imaging Eye exam

DOI

Accessible metadata

Indexed metadata

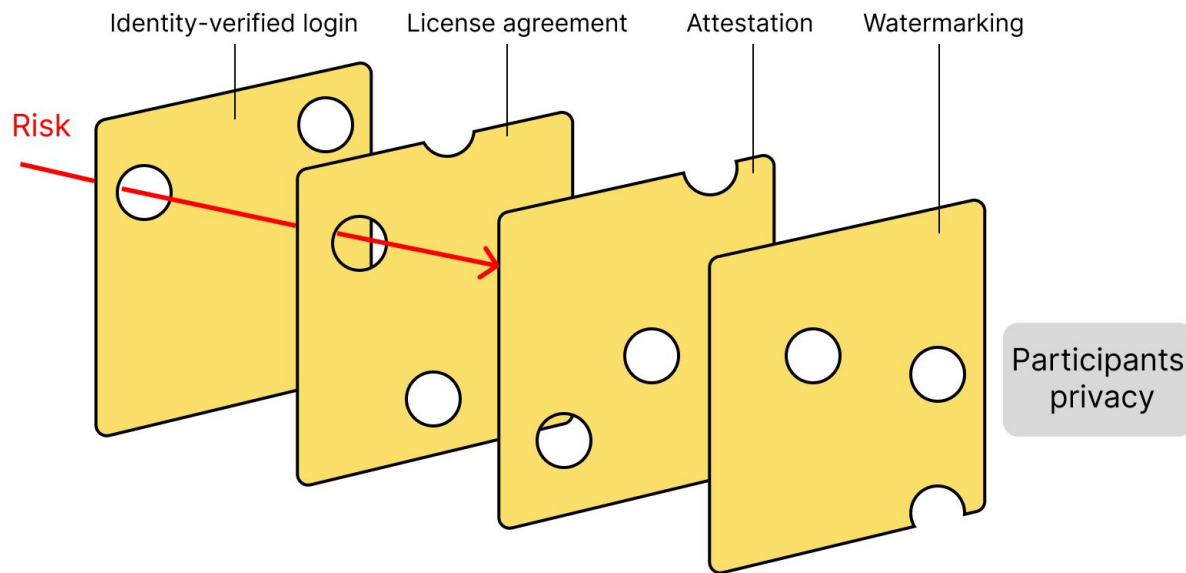
Clear access process

<https://doi.org/10.60775/fairhub.2>

The AI-READI Dataset

Beyond FAIR: Protecting participants privacy

Novel multiple-layered data access process on FAIRhub





Closing Remarks

Closing Remarks

Making data FAIR is not easy...

> 1,000 standards

> 1,600 databases

> 100 policies

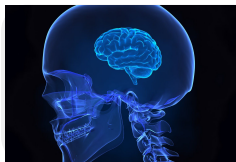


(“Bio” related results on FAIRsharing.org)

Closing Remarks

... but you can get started today with simple steps!

Kids



Share/register dataset on a suitable repository or registry

[NIH list of data repositories](#)

Adults



Provide as much metadata as possible

For example: authors, funding source, keywords, devices used, etc.

Legends



Structure (meta)data following standards

[fairsharing.org](#)

Closing Remarks

FAIR is good, AI-ready is better!

New Results

Follow this preprint

AI-readiness for Biomedical Data: Bridge2AI Recommendations

 Timothy Clark,  Harry Caufield,  Jillian A. Parker,  Sadnan Al Manir,  Edilberto Amorim,  James Eddy,  Nayoon Gim,  Brian Gow,  Wesley Goar,  Melissa Haendel,  Jan N. Hansen,  Nomi Harris,  Henning Hermjakob,  Marcin Joachimiak,  Gianna Jordan,  In-Hee Lee,  Shannon K. McWeeney,  Camille Nebeker,  Milen Nikolov,  Jamie Shaffer,  Nathan Sheffield,  Gloria Sheynkman,  James Stevenson,  Jake Y. Chen,  Chris Mungall,  Alex Wagner,  Sek Won Kong,  Satrajit S. Ghosh,  Bhavesh Patel,  Andrew Williams,  Monica C. Munoz-Torres the Bridge to Artificial Intelligence (Bridge2AI) Consortium members

doi: <https://doi.org/10.1101/2024.10.23.619844>

This article is a preprint and has not been certified by peer review [what does this mean?].

Abstract

Full Text

Info/History

Metrics

Preview PDF

Abstract

Biomedical research and clinical practice are in the midst of a transition toward significantly increased use of artificial intelligence (AI) and machine learning (ML) methods. These advances promise to enable qualitatively deeper insight into complex challenges formerly beyond the reach of analytic methods and human intuition while placing increased demands on ethical and explainable artificial intelligence (XAI), given the opaque nature of many deep learning methods.

<https://doi.org/10.1101/2024.10.23.619844>

Thank You!



bpatel@calmi2.org



fairdataihub.org

*Find these slides and
all resources here*



bit.ly/AI-ready